

5. Satellites can be positioned in orbit to communicate with base stations on the ground. Messages can be relayed around the world in this way. The diagram shows base stations **A**, **B**, **C** and **D** and four satellites.

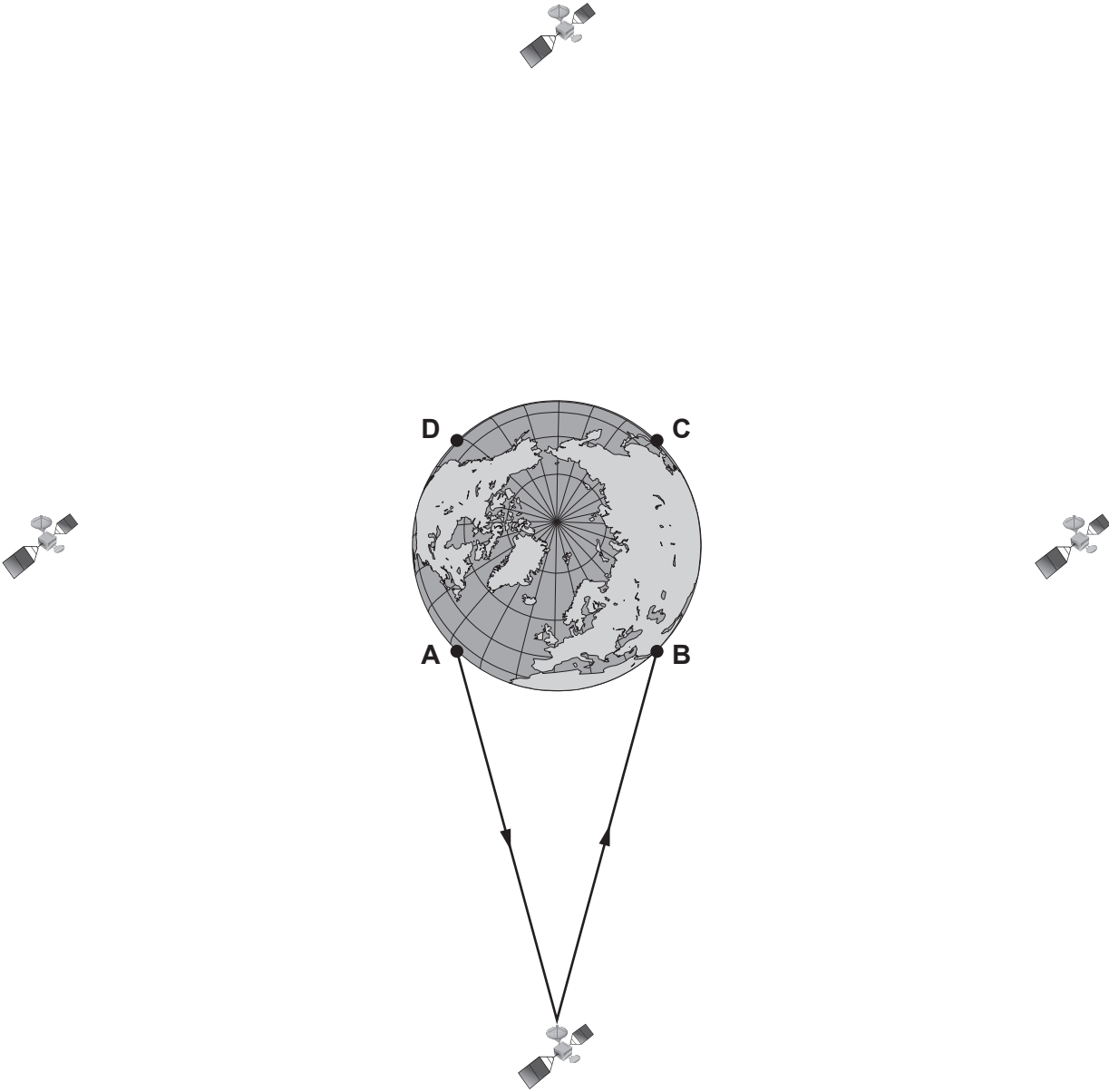


Diagram is not drawn to scale

- (a) The diagram shows the path of a signal sent from base station **A** to base station **B**. **Continue the path of the signal** so that it can be received at base station **C**. [1]



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- (b) The distance from a base station to a satellite is 36 000 km. Calculate the total distance travelled by the signal in going from **A** to **C**.
Give your answer in **metres**. [3]

Total distance = m

- (c) The time taken for the signal to travel from **A** to **C** is 0.48 s. Use an equation from page 2 to calculate the speed of the signal. [2]

Speed = m/s

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